

THE PROBLEM

Border CCTV is always recording. Human attention is not.

INTELLIGENT BORDER VIDEO ANALYTICS PLATFORM

Software • Surveillance & Security • Working prototype

24 × 7

Feeds never pause—even when operators must.

FINITE

Attention must scan many simultaneous scenes.

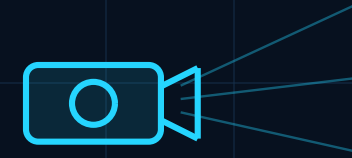
SECONDS

Brief movement can become an operational incident.

THE CORE FAILURE

A passive camera can explain yesterday.

It cannot guarantee that the right event gets attention now.

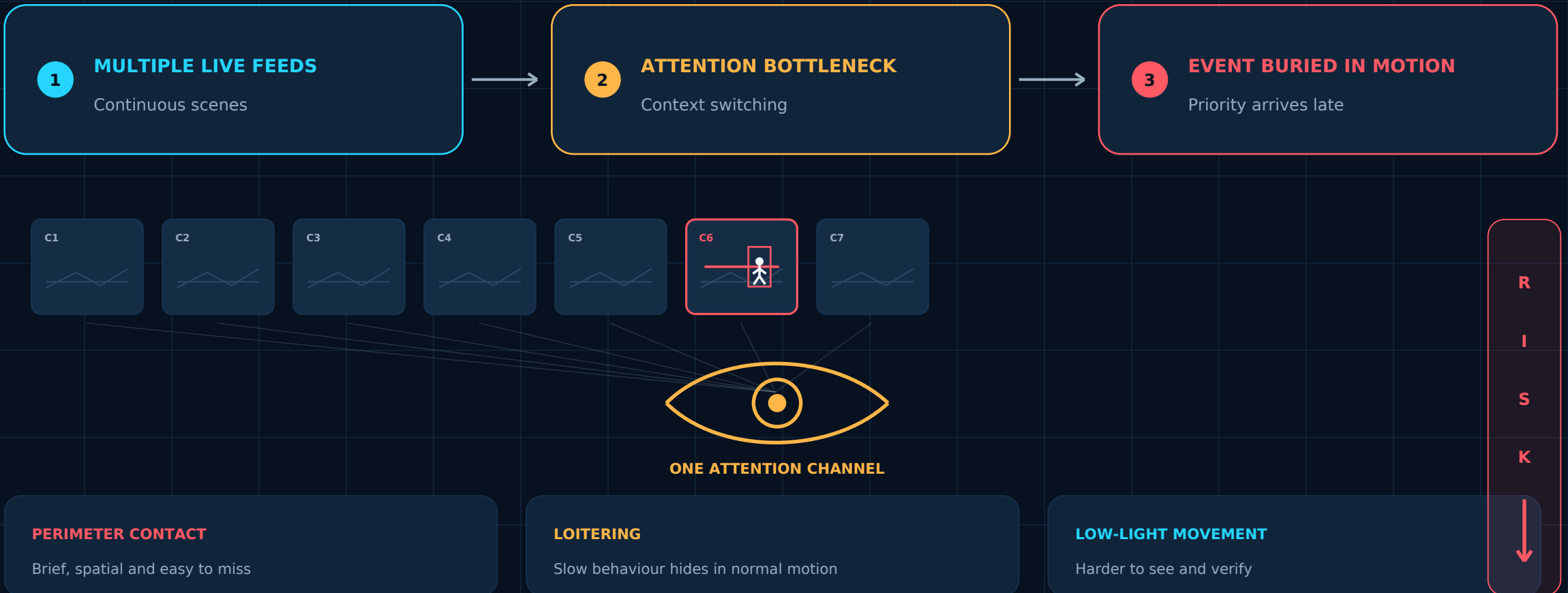


VISIBILITY ≠ AWARENESS

Recording is not recognition.

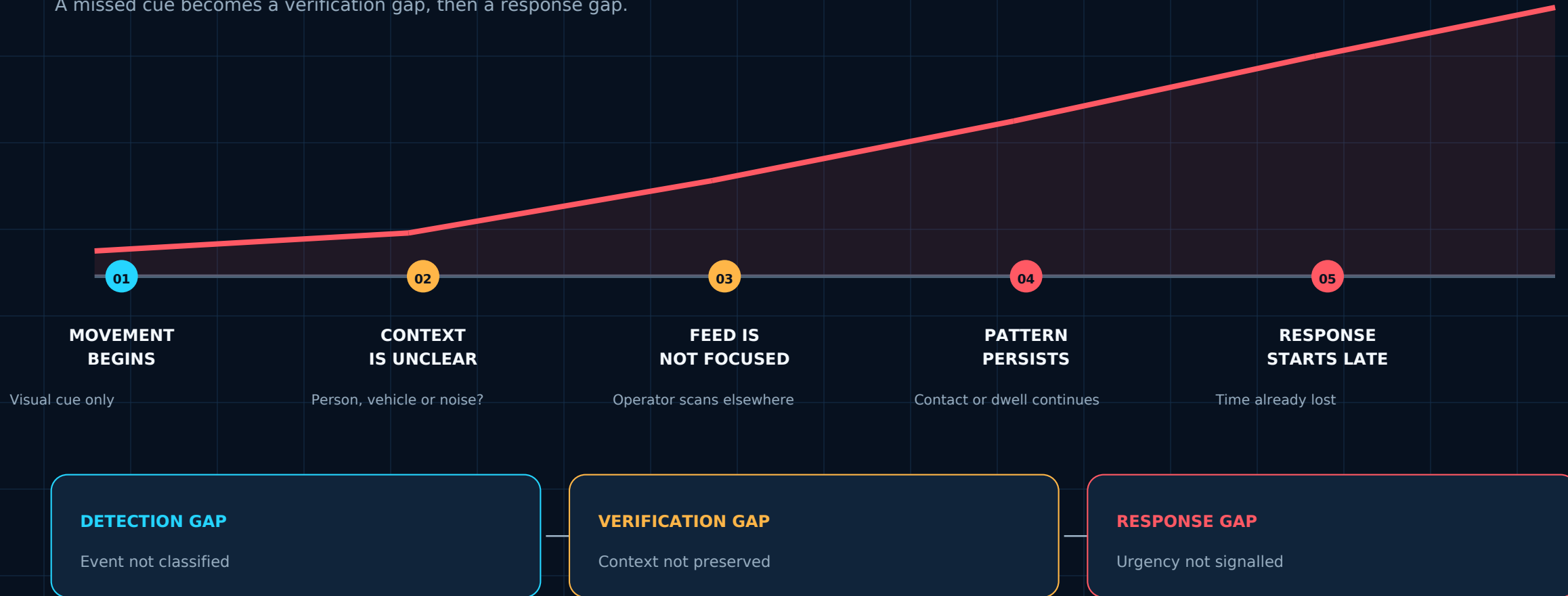
The blind spot is not visibility—it is attention.

Many feeds compete for one finite human decision channel.



Every delayed signal compounds operational risk.

A missed cue becomes a verification gap, then a response gap.

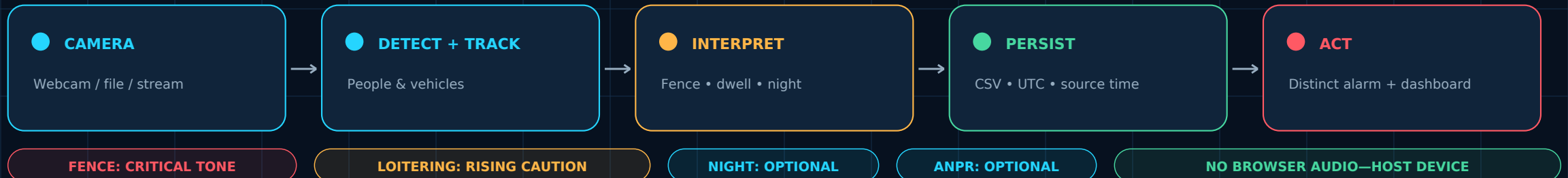
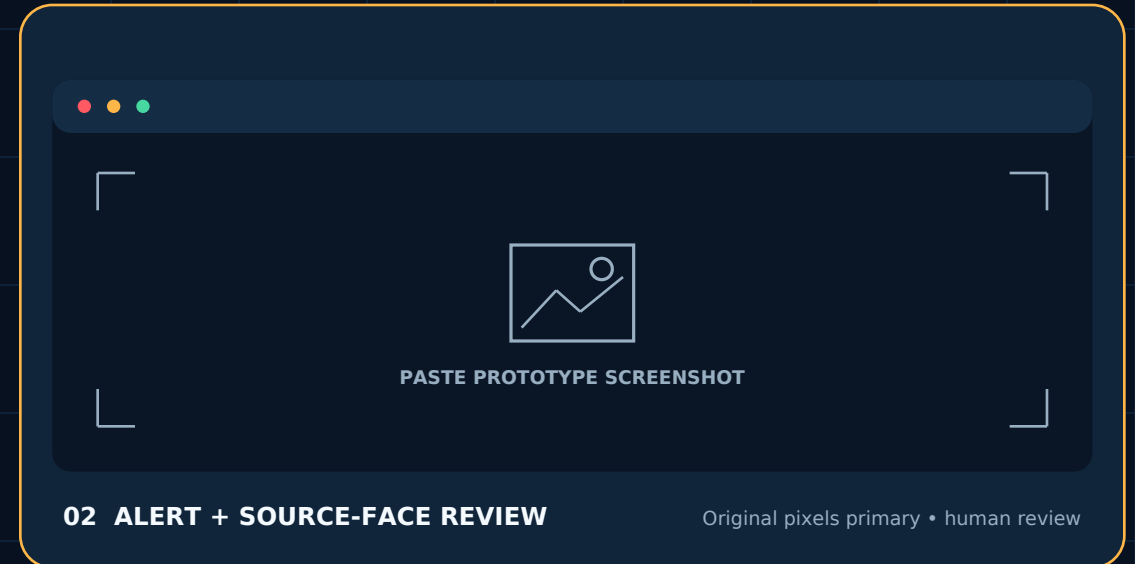
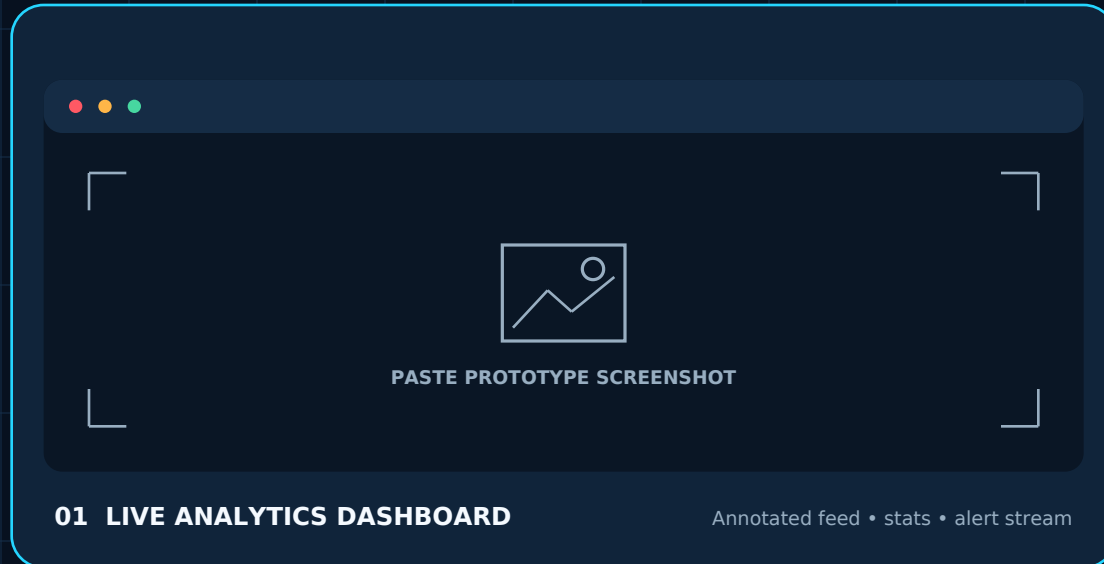


OPERATIONAL REQUIREMENT

Understand behaviour • preserve context • signal urgency • keep a human decision-maker in control

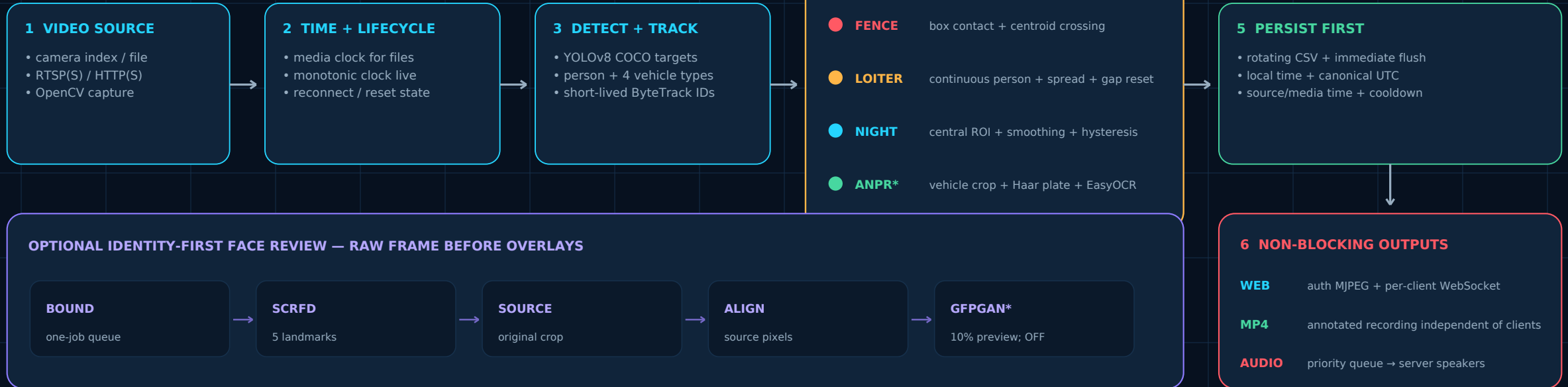
IBVAP turns one camera into an active watchpoint.

Implemented and validated in the latest published prototype • commit a571934



Pixels → events → evidence → action.

The capture path stays live; optional and slow work is bounded and failure-isolated.



SECURITY RAIL

HMAC HttpOnly session

host + origin checks

rate / connection caps

TLS-ready headers

rotating audit JSONL

model checksums

VALIDATED BUILD

74 tests

96.57% scoped coverage

Ruff + Bandit + 4 dependency audits

macOS OpenCV CI

* Optional and disabled by default. Coverage scope: utils/security; hardware and ML accuracy require representative footage.

What makes IBVAP different.

CAPABILITY	BASIC CCTV	SIMPLE MOTION	CLOUD-AI PATTERN	IBVAP PROTOTYPE
Understands behaviour	Recording only	Pixel change	Vendor-defined events	Fence contact + crossing • loiter • low-light • optional ANPR
Acts without WAN	Playback later	Generic buzzer	Often remote-dependent	Local analytics • MP4 • CSV • distinct host alarms
Delivers each alert	Manual review	Single signal	Remote notification	Persist-first • bounded history • queue per WebSocket client
Handles lifecycle	Recorder-specific	Basic state	Service-specific	Reset on reconnect/loop/time reversal • stale state expiry
Treats face evidence	Digital zoom	Not applicable	May enhance pixels	Original SCRFD crop is primary • generated preview separate/OFF
Shows assurance	Appliance-specific	Limited	Vendor-specific	Signed sessions • limits • audits • pinned locks • 74 tests

TECHNICAL BASIS

commit a571934 • Ultralytics YOLOv8 • ByteTrack • OpenCV • FastAPI/Starlette • InsightFace SCRFD

OPERATIONAL IMPACT

Earlier attention • differentiated urgency

Existing camera reuse • reviewable context

FEASIBLE NOW

Single camera • local or authenticated server

Python 3.11 • macOS / Linux / Windows

RESPONSIBLE BOUNDARY

Human verification mandatory • no identity/re-ID

2D calibration • prototype, not certified

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SIH SUBMISSION

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IMPORTANT INSTRUCTIONS

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Please ensure every
pointer is met before
submitting the Idea PDF.

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TEAM IBVAP

PLEASE ENSURE THE FOLLOWING:

1

Kindly keep the maximum slides limit up to six (6). (Including the title slide)

2

Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures

3

Keep your explanation precise and easy to understand

4

Idea should be unique and novel.

5

You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).

6

You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

NOTE

This instructions page may be deleted when uploading the idea details on the SIH portal.